

UI/UX Design Report

Final colour palette, visual rationale, accessibility and interaction design

Final Colour Palette

Colour	HEX	Primary use
Primary Dark Green	#005114	Primary actions, headings, borders
Accent Green	#34C759	Positive/active accents and highlights
Light Green	#E8FEEE	Navigation and supporting surfaces
Charcoal	#1F1F1F	Main text and icons
White	#FFFFFF	Main screen background
Heart Rate Pink	#FFE0E1	Heart-rate category
Blood Oxygen Blue	#E0ECFF	Blood-oxygen category
Steps Yellow	#FBFCC4	Steps/activity category
Active Energy Grey	#D5D2D4	Active-energy category
Sleep Green	#E8FEEE	Sleep category
Placeholder Grey	#D9D9D9	Placeholders/inactive visual elements

Primary visual system: #005114 (primary) → #34C759 (accent) → #E8FEEE (supporting background) → #FFFFFF (base)

Health-category system: pink = heart rate; blue = blood oxygen; yellow = steps; grey = active energy; light green = sleep.

1. Introduction

The final interface was designed as a connected health and wearable-monitoring application that allows users to view health information, connect wearable devices, manage patient information and communicate with healthcare-related users. The design prioritises clarity, consistency, accessibility and a calm visual experience. The final prototype uses a green-based visual identity supported by white space, dark text/icons and soft pastel colours for health-data categories. This combination was selected to create a trustworthy and approachable interface while keeping information visually organised.

2. Overall Visual Design Rationale

The visual design follows a simple and consistent mobile-app structure. Most screens use a white background with a light-green navigation area at the bottom. This creates a stable visual anchor across the prototype and

makes the three main destinations—Summary, Connect and Settings—easy to recognise. Dark icons and labels are used within the navigation bar so that the controls remain visible against the light background.

The interface uses rounded cards, moderate spacing and simple line icons to reduce visual complexity. The dashboard presents health information as separate cards, allowing users to scan individual measures without having to interpret a dense data table. The connection screens use large central icons and short messages such as “Scanning”, “No Devices yet!” and “Device connected successfully!” so that the user's current system state is immediately understandable.

A consistent layout was also maintained across patient and doctor/GP screens. Profile information, search results and editing functions use repeated patterns such as headings, back navigation, bordered content areas and confirmation screens. This consistency reduces the amount of new information users need to learn when moving between screens.

3. Final Colour Scheme

Green was selected as the main brand colour because the application is focused on health, wellbeing and connected monitoring. The primary dark green (#005114) is used for important headings, actions, borders and selected interface elements. It provides a strong visual anchor and creates a consistent identity throughout the prototype.

A brighter green (#34C759) is used as a secondary accent. It appears mainly in positive or active interface elements and decorative highlights rather than as the main colour for small text. The light green (#E8FEEE) is used for navigation backgrounds and supportive surfaces. This produces a softer version of the primary colour and prevents the interface from becoming visually heavy.

White (#FFFFFF) is the dominant screen background. It creates visual space around the information and allows the green controls and pastel health cards to stand out. Dark charcoal (#1F1F1F) is used for most icons and important text because it provides strong contrast and prevents the interface from depending too heavily on green.

The health dashboard uses a secondary pastel colour system. Heart rate uses soft pink (#FFE0E1), blood oxygen uses soft blue (#E0ECFF), steps use soft yellow (#FBFCC4), active energy uses soft grey (#D5D2D4), and sleep uses light green (#E8FEEE). These colours are intentionally low-saturation so that the dashboard remains calm while each category can be identified quickly.

4. Colour Accessibility

Accessibility was considered during the finalisation of the colour system. The design does not rely on colour as the only method of communicating information. For example, health metrics are differentiated through a combination of colour, icons and text labels. Heart rate is not identified only by pink; it also contains a heart icon and the “Heart rate” label. The same principle is applied to blood oxygen, steps, active energy and sleep. This is important for users with colour-vision deficiencies because people may perceive or distinguish colours differently. WCAG 2.2 Success Criterion 1.4.1 states that colour should not be the only visual means of communicating information (W3C, 2025).

Contrast was also considered when selecting the primary colours. WCAG 2.2 requires a minimum contrast ratio of 4.5:1 for normal-sized text and 3:1 for large text under Success Criterion 1.4.3. Meaningful non-text interface components and graphical objects should generally achieve at least 3:1 against adjacent colours under Success Criterion 1.4.11 (W3C, 2025). The final design therefore uses dark green (#005114) and dark charcoal (#1F1F1F) for important text and icons against light or white backgrounds. The brighter green (#34C759) is treated primarily as an accent rather than as small body text on white, because its contrast is lower.

The use of icons and labels alongside colour also means that the pastel dashboard cards remain understandable if users cannot distinguish every colour difference. This follows the accessibility principle that colour coding can be useful when it is supported by additional visual or textual cues.

5. Typography

Typography was selected to create a friendly and approachable visual identity while maintaining hierarchy. The prototype uses a rounded display style, including Coiny in prominent interface text, which gives the application a distinctive and informal character. Larger and bolder headings are used to establish screen hierarchy, while smaller text is used for supporting information, labels and status messages.

Typography is also combined with spacing rather than relying only on size. Headings are positioned consistently near the top of screens, while cards and navigation labels have clear separation. This helps users scan the interface and understand which information belongs together.

6. Interaction and Navigation

The prototype uses a persistent bottom navigation structure with three main areas: Summary, Connect and Settings. Keeping these controls in a consistent location reduces navigation effort and supports recognition across screens. The Connect area provides states for an empty device list, Bluetooth-disabled status, device scanning, device discovery, successful connection and failed connection. Presenting these states separately makes system feedback visible and gives users an appropriate next action, such as opening settings, rescanning or trying again.

The patient/doctor functionality also follows a clear sequence. Users can search for a patient, open a patient profile, edit information, review the proposed changes and receive confirmation after saving. The confirmation screen provides explicit feedback that the action has been completed. These interaction patterns were designed to reduce uncertainty and make system status visible.

7. Consistency and Visual Hierarchy

Consistency was maintained through repeated use of the same green palette, rounded cards, icon style, navigation structure and spacing. The primary dark green identifies important actions and headings, while light green and pastel surfaces distinguish supporting content. Dark icons provide a consistent functional language.

Visual hierarchy is created through size, contrast, position and grouping. Important headings are more prominent than supporting labels. Primary actions such as “Rescan”, “Open setting”, “Save” and confirmation actions are visually emphasised. Secondary information is placed within lighter cards or smaller text. This hierarchy helps users identify what they should look at or do next without requiring every element to have the same visual weight.

8. Design Reflection

The final design balances a health-oriented visual identity with the practical requirements of a mobile monitoring application. The strongest design decision is the controlled use of green as a consistent brand colour rather than applying many unrelated strong colours throughout the interface. The pastel colours are reserved for categorising health metrics, while dark green and charcoal are used for functional information.

An important accessibility decision was to avoid making colour the sole indicator of status or meaning. Although red and green are used in some device-status cards, the interface also provides text, icons and

numerical information. If the prototype were developed further, additional status words such as “Connected”, “Low”, “Warning” or “Normal” could be added to status cards to make the distinction even more explicit.

Overall, the final UI aims to be visually calm, recognisable and easy to scan. The combination of a consistent colour system, repeated navigation patterns, clear iconography and supporting text creates a cohesive experience across the application's major functions.

Accessibility Contrast Checks

The following checks are included to document the intended use of the palette. The dark green and charcoal colours are reserved for high-priority text and icons; the brighter green is not intended for small body text on white.

Foreground / Background	Approx. contrast	Design use
#005114 / #FFFFFF	9.60:1	Primary text, buttons, headings
#1F1F1F / #FFFFFF	16.48:1	Main text and icons
#005114 / #E8FEEE	9.07:1	Green text/icons on light green
#34C759 / #FFFFFF	2.22:1	Accent only; avoid small text

References

World Wide Web Consortium (W3C). (2025). Understanding Success Criterion 1.4.1: Use of Color (WCAG 2.2). <https://www.w3.org/WAI/WCAG22/Understanding/use-of-color>

World Wide Web Consortium (W3C). (2025). Understanding Success Criterion 1.4.3: Contrast (Minimum). <https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum>

World Wide Web Consortium (W3C). (2025). Understanding Success Criterion 1.4.11: Non-text Contrast (WCAG 2.2). <https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast>